Automated Model Risk Auditor | Architecture

The system is split into components with one job each: a suite manager, a dual-test engine, an attack generator, an isolated sandbox, a judge orchestra with a human queue, a risk engine, a signed report service and a post-release monitor.

3.1 Overview: components, roles, interfaces

Component	Plane	Role	Boundary
Suite manager	Test pipeline	Keeps every test in versions, including the ones the generator invents. Hands each run a batch.	Inside
Dual-test engine	Test pipeline	Sends every test twice: once as a declared test (Mode A), once as an ordinary conversation (Mode B). Compares the answers.	Inside
Model under test	Test pipeline	The model submitted for testing. Outside the system, and not allowed to learn about it.	External
Attack generator	Adversarial research	Invents new attack phrasings with three methods, like a real attacker, within a fixed budget.	Inside
Sandbox coordinator	Adversarial research	Runs dangerous tests in an environment that looks real but is cut off from the world, and deletes it after each test.	Inside
Judge orchestra	Judging and decision	Passes every answer to three different judges (rules and two models) and computes their agreement.	Inside
Human review queue	Judging and decision	Cases where the judges disagreed wait here. A person decides.	Inside
Risk engine	Judging and decision	Computes a score per category, applies a hard threshold to critical categories, compares with earlier versions.	Inside
Report service	Reporting and follow-up	Assembles results into a three-level report, produces a PDF and signs it.	Inside

Component	Plane	Role	Boundary
Post-release monitor	Reporting and follow-up	Samples real use of the model, sends it to the judges, compares with what was tested before release. Detects drift.	Inside
Key management	Cross-cutting layer	Holds the key used to sign the report.	Inside
External stop mechanism	Reporting and follow-up	A separate project the monitor asks when it wants a model suspended. Outside the system and not under its control.	External

Links

- Attack generator → Suite manager
- Suite manager → Dual-test engine
- Dual-test engine → Model under test
- Dual-test engine → Sandbox coordinator
- Sandbox coordinator → Model under test
- Dual-test engine → Judge orchestra
- Judge orchestra → Human review queue
- Judge orchestra → Risk engine
- Risk engine → Report service
- Post-release monitor → Judge orchestra
- Post-release monitor → External stop mechanism
- Key management → Report service

3.2 Mapping to the spec

#	Feature	Components
1	Managing the test suite	Suite manager, Dual-test engine
2	Detecting behavior change	Dual-test engine, Judge orchestra
3	Three judges	Judge orchestra, Human review queue
4	Score and release gate	Risk engine
5	Generating attack tests	Attack generator, Suite manager
6	Running in an isolated room	Sandbox coordinator, Model under test
7	Regression tracking	Risk engine, Suite manager
8	Signed report	Report service, Key management
9	Post-release monitoring	Post-release monitor, Judge orchestra, External stop mechanism

3.3 Data design

Transient

- **TestCase:** One test with its category, wording and expected behavior.
- **ModePair:** The same test in its two framings (declared and ordinary) with both answers.
- **JudgeVerdict:** One judge's opinion on one answer, with its reasoning.

Persisted

- **SuiteVersion:** A frozen version of the whole test suite, so results stay comparable.
- **RunRecord:** What ran, against which model, when, with which suite version.
- **JudgeLog:** Every judge's output, kept for audit and for human reviewers.
- **SignedReport:** The final report with its signature. The official artifact.
- **DriftSample:** A sample of real traffic used to compare behavior after release.

Architecture decisions

- **One big service, or separate components?** Chosen: Separate components
- **Run dangerous tests inside the system, or in a separate room?** Chosen: A separate room
- **Judges in sequence or in parallel?** Chosen: In parallel
- **Should the system suspend a model itself, or ask someone else?** Chosen: Ask an external mechanism

Build order

- Sprint 1: a working skeleton (one test, end to end: question, answer, one judge, score, simple report)
- Sprint 2: real judging (three judges, agreement measure, human queue)
- Sprint 3: isolated room and attack tests (attack generator, dangerous tests in the room)
- Sprint 4: signed report (three levels, PDF and signature, comparison with earlier versions)
- Sprint 5: post-release follow-up (field sampling, drift detection, suspension request)

Closure test

Open loop. Two elements sit outside the boundary: the model under test and the external stop mechanism. The monitor detects drift but can only send a request to a mechanism it does not control. Detection without internal enforcement leaves the loop open.